

Home Work 2
II B. Tech. (Common)
Mathematics-II (MCI102)

Notations:

$M_{m \times n}(\mathbb{F})$: The set of all matrices of order $m \times n$ with entries from $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.

$P_n(\mathbb{R})$ or $\mathbb{R}_n[x]$: The set of all polynomials of degree at most n in one variable with real coefficients.

$P(\mathbb{R})$ or $\mathbb{R}[x]$: The set of all polynomials in one variable with real coefficients.

$C(\mathbb{R})$: The set of all real-valued continuous functions

$\mathbb{R}^\infty$: The set of all real-valued sequences.

$\mathbb{R}^+$: The set of positive real numbers.

1. Which of the following are vector spaces?

(a) $V = \mathbb{R}^2$ over $\mathbb{R}$ with componentwise addition and $\lambda \odot (x, y) = (3\lambda x, y)$ for all $(x, y) \in \mathbb{R}^2$ and $\lambda \in \mathbb{R}$.

(b) $V = \mathbb{R}^+$ over $\mathbb{R}$ with $x \oplus y = xy$ and $\lambda \odot x = x^\lambda$ for all $x, y \in \mathbb{R}^+$ and $\lambda \in \mathbb{R}$.

(c) $V = \mathbb{R}[x]$ over $\mathbb{R}$ with usual addition and scalar multiplication of polynomials.

(d) $V = \mathbb{R}^\infty$ over $\mathbb{R}$ with $a + b = \{a_n + b_n\}_{n=1}^\infty$ and $\lambda a = \{\lambda a_n\}_{n=1}^\infty$ for all $a = \{a_n\}_{n=1}^\infty$, $b = \{b_n\}_{n=1}^\infty \in \mathbb{R}^\infty$ and $\lambda \in \mathbb{R}$.

(e) $V = \{t_\alpha : \mathbb{R} \rightarrow \mathbb{R} \mid t_\alpha(x) = x + \alpha, \alpha \in \mathbb{R}\}$ over $\mathbb{R}$ with composition of mappings and $\lambda t_\alpha = t_{\alpha\lambda}$ for all $t_\alpha \in V$ and $\lambda \in \mathbb{R}$.

(f) $V = C[a, b]$ over $\mathbb{R}$ with $(f + g)(x) = f(x) + g(x)$ and $(\lambda \cdot f)(x) = \lambda f(x)$ for all $\lambda \in \mathbb{R}, f, g \in C[a, b]$.

(g) V is the set of all real-valued continuous functions defined on an open interval I which have at most finite number of points of discontinuity over $\mathbb{R}$ with pointwise addition and scalar multiplication of functions.

(h) $V = \{\text{all real polynomials of degree 4 or 6}\}$ over $\mathbb{R}$ with usual addition and scalar multiplication.

(i) $V = \{\text{all } n \times n \text{ Hermitian matrices}\}$ over $\mathbb{C}$ with usual addition and scalar multiplication.

(j) $V = \{\text{all } n \times n \text{ skew-Hermitian matrices}\}$ over $\mathbb{C}$ with usual matrix addition and scalar multiplication.

Answers: (a) No (b) Yes (c) Yes (d) Yes (e) Yes (f) Yes (g) Yes (h) No (i) No (j) No.

Solutions: (a) To determine whether V is a vector space over $\mathbb{R}$, we verify the vector space axioms. In particular, the identity axiom for scalar multiplication requires that

$$1 \odot (x, y) = (x, y) \quad \text{for all } (x, y) \in V.$$

Using the given scalar multiplication,

$$1 \odot (x, y) = (3 \cdot 1 \cdot x, y) = (3x, y).$$

Since $(3x, y) \neq (x, y)$ for all $(x, y) \in \mathbb{R}^2$ (for example, when $x \neq 0$), the identity axiom is not satisfied.

Therefore, V with the given operations is not a vector space over $\mathbb{R}$.

(b) We verify the vector space axioms one by one.

(A1) Closure under addition. For $x, y \in V$,

$$x \oplus y = xy > 0, \text{ thus, } x \oplus y \in V.$$

(A2) Associativity of addition. For $x, y, z \in V$,

$$(x \oplus y) \oplus z = (xy)z = x(yz) = x \oplus (y \oplus z).$$

(A3) Additive identity. We seek $0_V \in V$ such that $x \oplus 0_V = x$ for all $x \in V$. Since $x0_V = x$, it follows that $0_V = 1$.

(A4) Additive inverse. For $x \in V$, the element $x^{-1} \in V$ satisfies

$$x \oplus x^{-1} = xx^{-1} = 1 = 0_V.$$

(A5) Commutativity of addition. For $x, y \in V$,

$$x \oplus y = xy = yx = y \oplus x.$$

(S1) Closure under scalar multiplication. For $\lambda \in \mathbb{R}$ and $x \in V$, $x^\lambda > 0$. Hence $\lambda \odot x \in V$.

(S2) Associativity of scalar multiplication. For $\lambda, \mu \in \mathbb{R}$ and $x \in V$,

$$(\lambda\mu) \odot x = x^{\lambda\mu} = (x^\mu)^\lambda = \lambda \odot (\mu \odot x).$$

(S3) Identity element of scalar multiplication. For $x \in V$,

$$1 \odot x = x^1 = x.$$

(D1) Distributivity over vector addition. For $\lambda \in \mathbb{R}$ and $x, y \in V$,

$$\lambda \odot (x \oplus y) = (xy)^\lambda = x^\lambda y^\lambda = (\lambda \odot x) \oplus (\lambda \odot y).$$

(D2) Distributivity over scalar addition. For $\lambda, \mu \in \mathbb{R}$ and $x \in V$,

$$(\lambda + \mu) \odot x = x^{\lambda+\mu} = x^\lambda x^\mu = (\lambda \odot x) \oplus (\mu \odot x).$$

Since all vector space axioms are satisfied, we conclude that

$\mathbb{R}^+$ with the operations $\oplus$ and $\odot$ is a vector space over $\mathbb{R}$.

(c) We verify the vector space axioms one by one.

(A1) Closure under addition. For $p(x), q(x) \in V$, their sum $p(x) + q(x)$ is also a polynomial, so $p(x) \oplus q(x) = p(x) + q(x) \in V$.

(A2) Associativity of addition. For $p(x), q(x), r(x) \in V$,

$$(p(x) \oplus q(x)) \oplus r(x) = (p(x) + q(x)) + r(x) = p(x) + (q(x) + r(x)) = p(x) \oplus (q(x) \oplus r(x)).$$

(A3) Additive identity. The zero polynomial $0 \in V$ satisfies

$$p(x) \oplus 0 = p(x) + 0 = p(x)$$

for all $p(x) \in V$.

(A4) Additive inverse. For $p(x) \in V$, the polynomial $-p(x) \in V$ satisfies

$$p(x) \oplus (-p(x)) = p(x) + (-p(x)) = 0.$$

(A5) Commutativity of addition. For $p(x), q(x) \in V$,

$$p(x) \oplus q(x) = p(x) + q(x) = q(x) + p(x) = q(x) \oplus p(x).$$

(S1) Closure under scalar multiplication. For $\lambda \in \mathbb{R}$ and $p(x) \in V$, $\lambda \odot p(x) = \lambda p(x)$ is also a polynomial, so $\lambda \odot p(x) \in V$.

(S2) Associativity of scalar multiplication. For $\lambda, \mu \in \mathbb{R}$ and $p(x) \in V$,

$$(\lambda\mu) \odot p(x) = (\lambda\mu)p(x) = \lambda(\mu p(x)) = \lambda \odot (\mu \odot p(x)).$$

(S3) Identity element of scalar multiplication. For $p(x) \in V$,

$$1 \odot p(x) = 1 \cdot p(x) = p(x).$$

(D1) Distributivity over vector addition. For $\lambda \in \mathbb{R}$ and $p(x), q(x) \in V$,

$$\lambda \odot (p(x) \oplus q(x)) = \lambda(p(x) + q(x)) = \lambda p(x) + \lambda q(x) = (\lambda \odot p(x)) \oplus (\lambda \odot q(x)).$$

(D2) Distributivity over scalar addition. For $\lambda, \mu \in \mathbb{R}$ and $p(x) \in V$,

$$(\lambda + \mu) \odot p(x) = (\lambda + \mu)p(x) = \lambda p(x) + \mu p(x) = (\lambda \odot p(x)) \oplus (\mu \odot p(x)).$$

Since all vector space axioms are satisfied, we conclude that

$\mathbb{R}[x]$ with the usual addition and scalar multiplication is a vector space over $\mathbb{R}$.

(d) We check the vector space axioms one by one.

(A1) Closure under addition. For $a, b \in V$, the sequence

$$a + b = \{a_n + b_n\}_{n=1}^{\infty}$$

is again a sequence of real numbers, so $a + b \in V$.

(A2) Associativity of addition. For $a, b, c \in V$,

$$(a + b) + c = \{(a_n + b_n) + c_n\}_{n=1}^{\infty} = \{a_n + (b_n + c_n)\}_{n=1}^{\infty} = a + (b + c).$$

(A3) Additive identity. The zero sequence

$$0 = \{0, 0, 0, \dots\} \in V$$

satisfies

$$a + 0 = \{a_n + 0\}_{n=1}^{\infty} = a$$

for all $a \in V$.

(A4) Additive inverse. For $a = \{a_n\}_{n=1}^{\infty} \in V$, the sequence

$$-a = \{-a_n\}_{n=1}^{\infty} \in V$$

satisfies

$$a + (-a) = \{a_n + (-a_n)\}_{n=1}^{\infty} = 0.$$

(A5) Commutativity of addition. For $a, b \in V$,

$$a + b = \{a_n + b_n\}_{n=1}^{\infty} = \{b_n + a_n\}_{n=1}^{\infty} = b + a.$$

(S1) Closure under scalar multiplication. For $\lambda \in \mathbb{R}$ and $a \in V$,

$$\lambda a = \{\lambda a_n\}_{n=1}^{\infty} \in V$$

since each component is a real number.

(S2) Associativity of scalar multiplication. For $\lambda, \mu \in \mathbb{R}$ and $a \in V$,

$$(\lambda\mu)a = \{(\lambda\mu)a_n\}_{n=1}^{\infty} = \{\lambda(\mu a_n)\}_{n=1}^{\infty} = \lambda(\mu a).$$

(S3) Identity element of scalar multiplication. For $a \in V$,

$$1 \cdot a = \{1 \cdot a_n\}_{n=1}^{\infty} = a.$$

(D1) Distributivity over addition of vectors. For $\lambda \in \mathbb{R}$ and $a, b \in V$,

$$\lambda(a + b) = \{\lambda(a_n + b_n)\}_{n=1}^{\infty} = \{\lambda a_n + \lambda b_n\}_{n=1}^{\infty} = \lambda a + \lambda b.$$

(D2) Distributivity over addition of scalars. For $\lambda, \mu \in \mathbb{R}$ and $a \in V$,

$$(\lambda + \mu)a = \{(\lambda + \mu)a_n\}_{n=1}^{\infty} = \{\lambda a_n + \mu a_n\}_{n=1}^{\infty} = \lambda a + \mu a.$$

Since all vector space axioms are satisfied, we conclude that

$\mathbb{R}^{\infty}$ with componentwise addition and scalar multiplication is a vector space over $\mathbb{R}$.

(e) We check the vector space axioms.

(A1) Closure under addition. For $t_{\alpha}, t_{\beta} \in V$,

$$(t_{\alpha} \circ t_{\beta})(x) = t_{\alpha}(t_{\beta}(x)) = t_{\alpha}(x + \beta) = x + \beta + \alpha = x + (\alpha + \beta).$$

This is of the form $t_{\gamma}(x) = x + \gamma$ with $\gamma = \alpha + \beta \in \mathbb{R}$, so $t_{\alpha} + t_{\beta} \in V$.

(A2) Associativity of addition. For $t_{\alpha}, t_{\beta}, t_{\gamma} \in V$,

$$(t_{\alpha} \circ t_{\beta}) \circ t_{\gamma} = t_{\alpha} \circ (t_{\beta} \circ t_{\gamma}),$$

because composition of functions is associative.

(A3) Additive identity. We need $0_V \in V$ such that $0_V \circ t_{\alpha} = t_{\alpha} \circ 0_V = t_{\alpha}$. Let $0_V = t_0$, then

$$t_0 \circ t_{\alpha}(x) = t_0(x + \alpha) = x + \alpha, \quad t_{\alpha} \circ t_0(x) = t_{\alpha}(x + 0) = x + \alpha,$$

so t_0 is the additive identity.

(A4) Additive inverse. For $t_{\alpha} \in V$, the function $t_{-\alpha} \in V$ satisfies

$$t_{\alpha} \circ t_{-\alpha}(x) = t_{\alpha}(x - \alpha) = x = t_0(x),$$

so $t_{-\alpha}$ is the additive inverse.

(A5) Commutativity of addition. For $t_{\alpha}, t_{\beta} \in V$,

$$t_{\alpha} \circ t_{\beta}(x) = x + \beta + \alpha = x + (\alpha + \beta) = x + (\beta + \alpha) = t_{\beta} \circ t_{\alpha}(x),$$

so addition is commutative.

(S1) Closure under scalar multiplication. For $\lambda \in \mathbb{R}$ and $t_{\alpha} \in V$,

$$\lambda t_{\alpha} = t_{\lambda\alpha} \in V,$$

so closure holds.

(S2) Associativity of scalar multiplication. For $\lambda, \mu \in \mathbb{R}$ and $t_{\alpha} \in V$,

$$(\lambda\mu)t_{\alpha} = t_{(\lambda\mu)\alpha} = t_{\lambda(\mu\alpha)} = \lambda(\mu t_{\alpha}),$$

so associativity holds.

(S3) Identity element of scalar multiplication. For $t_{\alpha} \in V$,

$$1 \cdot t_{\alpha} = t_{1\alpha} = t_{\alpha},$$

so the scalar identity holds.

(D1) Distributivity over addition of vectors. For $\lambda \in \mathbb{R}$ and $t_{\alpha}, t_{\beta} \in V$,

$$\lambda(t_{\alpha} + t_{\beta}) = \lambda(t_{\alpha} \circ t_{\beta}) = \lambda t_{\alpha+\beta} = t_{\lambda(\alpha+\beta)}.$$

On the other hand,

$$\lambda t_{\alpha} + \lambda t_{\beta} = t_{\lambda\alpha} \circ t_{\lambda\beta} = t_{\lambda\alpha+\lambda\beta} = t_{\lambda(\alpha+\beta)}.$$

Thus, distributivity holds.

(D2) Distributivity over addition of scalars. For $\lambda, \mu \in \mathbb{R}$ and $t_\alpha \in V$,

$$(\lambda + \mu)t_\alpha = t_{(\lambda+\mu)\alpha} = t_{\lambda\alpha+\mu\alpha} = t_{\lambda\alpha} \circ t_{\mu\alpha} = \lambda t_\alpha + \mu t_\alpha,$$

so this distributive law holds.

Since all vector space axioms are satisfied, we conclude that

$$V = \{t_\alpha(x) = x + \alpha \mid \alpha \in \mathbb{R}\} \text{ is a vector space over } \mathbb{R}.$$

(f)

We verify the vector space axioms.

(A1) Closure under addition. For $f, g \in C[a, b]$, the function $f + g$ is continuous on $[a, b]$, so $f + g \in C[a, b]$.

(A2) Associativity of addition. For $f, g, h \in C[a, b]$,

$$(f + g) + h = f + (g + h)$$

because addition of real numbers is associative for each $x \in [a, b]$.

(A3) Additive identity. The zero function $0(x) = 0$ for all $x \in [a, b]$ satisfies

$$f + 0 = f$$

for all $f \in C[a, b]$.

(A4) Additive inverse. For $f \in C[a, b]$, the function $-f \in C[a, b]$ satisfies

$$f + (-f) = 0.$$

(A5) Commutativity of addition. For $f, g \in C[a, b]$,

$$f + g = g + f$$

since real addition is commutative for each $x \in [a, b]$.

(S1) Closure under scalar multiplication. For $\lambda \in \mathbb{R}$ and $f \in C[a, b]$, λf is continuous on $[a, b]$, so $\lambda f \in C[a, b]$.

(S2) Associativity of scalar multiplication. For $\lambda, \mu \in \mathbb{R}$ and $f \in C[a, b]$,

$$(\lambda\mu)f = \lambda(\mu f)$$

pointwise for all $x \in [a, b]$.

(S3) Identity element of scalar multiplication. For $f \in C[a, b]$,

$$1 \cdot f = f.$$

(D1) Distributivity over addition of functions. For $\lambda \in \mathbb{R}$ and $f, g \in C[a, b]$,

$$\lambda(f + g) = \lambda f + \lambda g$$

pointwise for all $x \in [a, b]$.

(D2) Distributivity over addition of scalars. For $\lambda, \mu \in \mathbb{R}$ and $f \in C[a, b]$,

$$(\lambda + \mu)f = \lambda f + \mu f$$

pointwise for all $x \in [a, b]$.

Since all vector space axioms are satisfied, we conclude that

$$C[a, b] \text{ with pointwise addition and scalar multiplication is a vector space over } \mathbb{R}.$$

(g) We verify the vector space axioms.

(A1) Closure under addition. If f and g have at most finitely many points of discontinuity, then $f + g$ has discontinuities only at points where f or g is discontinuous. Therefore, $f + g$ has at most finitely many points of discontinuity, so $f + g \in V$.

(A2) Associativity of addition. For $f, g, h \in V$,

$$(f + g) + h = f + (g + h)$$

pointwise, because addition of real numbers is associative.

(A3) Additive identity. The zero function $0(x) = 0$ for all $x \in I$ belongs to V (it has no discontinuities) and satisfies

$$f + 0 = f$$

for all $f \in V$.

(A4) Additive inverse. For $f \in V$, the function $-f$ belongs to V (it has the same points of discontinuity as f) and satisfies

$$f + (-f) = 0.$$

(A5) Commutativity of addition. For $f, g \in V$,

$$f + g = g + f$$

pointwise.

(S1) Closure under scalar multiplication. For $\lambda \in \mathbb{R}$ and $f \in V$, the function λf has discontinuities only at the same points as f , which are finitely many. Hence $\lambda f \in V$.

(S2) Associativity of scalar multiplication. For $\lambda, \mu \in \mathbb{R}$ and $f \in V$,

$$(\lambda\mu)f = \lambda(\mu f)$$

pointwise.

(S3) Identity element of scalar multiplication. For $f \in V$,

$$1 \cdot f = f.$$

(D1) Distributivity over addition of functions. For $\lambda \in \mathbb{R}$ and $f, g \in V$,

$$\lambda(f + g) = \lambda f + \lambda g$$

pointwise.

(D2) Distributivity over addition of scalars. For $\lambda, \mu \in \mathbb{R}$ and $f \in V$,

$$(\lambda + \mu)f = \lambda f + \mu f$$

pointwise.

Since all vector space axioms are satisfied, we conclude that

V with pointwise addition and scalar multiplication is a vector space over $\mathbb{R}$.

(h) We check the vector space axioms.

Closure under addition. Take $p(x), q(x) \in V$. Then $\deg(p) = 4$ or 6 and $\deg(q) = 4$ or 6 . If $\deg(p) = \deg(q) = 4$, then $\deg(p + q) \leq 4$. If the leading coefficients cancel, $\deg(p + q) < 4$, which is not in V . Hence, V is not closed under addition. Since closure under addition fails, V does not satisfy the vector space axioms. Therefore,

V is **not** a vector space over $\mathbb{R}$.

(i) We verify the vector space axioms over $\mathbb{C}$.

(A1) Closure under addition. If $A, B \in V$, then

$$(A + B)^* = A^* + B^* = A + B,$$

so $A + B \in V$.

(A2) Associativity of addition. Matrix addition is associative:

$$(A + B) + C = A + (B + C)$$

for all $A, B, C \in V$.

(A3) Additive identity. The zero matrix 0 satisfies $0^* = 0$ and $A + 0 = A$ for all $A \in V$.

(A4) Additive inverse. For $A \in V$, $-A \in V$ since $(-A)^* = -A^* = -A$, and $A + (-A) = 0$.

(A5) Commutativity of addition. Matrix addition is commutative:

$$A + B = B + A$$

for all $A, B \in V$.

(S1) Closure under scalar multiplication. If $A \in V$ and $\lambda \in \mathbb{C}$, then

$$(\lambda A)^* = \bar{\lambda} A^* = \bar{\lambda} A.$$

However, for Hermitian matrices, closure under all complex scalars fails because $\bar{\lambda} A \neq \lambda A$ in general unless $\lambda \in \mathbb{R}$.

Conclusion: V is not a vector space over $\mathbb{C}$.

Remark. If we consider V over $\mathbb{R}$, all vector space axioms hold, because for $\lambda \in \mathbb{R}$:

$$(\lambda A)^* = \lambda A^* = \lambda A,$$

and all other axioms are satisfied. Over $\mathbb{C}$, scalar multiplication fails to preserve Hermitian property, so V is not a vector space over $\mathbb{C}$ but V is a vector space over $\mathbb{R}$.

(j). Following as the above, V is not a vector space over $\mathbb{C}$ but V is a vector space over $\mathbb{R}$.

2. On $\mathbb{R}^2$, define two operations

$$(x_1, y_1) \oplus (x_2, y_2) = (x_1 - x_2, y_1 - y_2),$$

$$c \odot (x, y) = (-cx, -cy)$$

where $c \in \mathbb{R}$. Which of the axioms for a vector space are satisfied by $(\mathbb{R}^2, \oplus, \odot)$ over $\mathbb{R}$?

Solution: We check the vector space axioms:

- **Closure under addition:** For any $(x_1, y_1), (x_2, y_2) \in \mathbb{R}^2$,

$$(x_1, y_1) \oplus (x_2, y_2) = (x_1 - x_2, y_1 - y_2) \in \mathbb{R}^2.$$

Satisfied.

- **Associativity of addition:** Let $(x_1, y_1), (x_2, y_2), (x_3, y_3) \in \mathbb{R}^2$.

$$((x_1, y_1) \oplus (x_2, y_2)) \oplus (x_3, y_3) = (x_1 - x_2 - x_3, y_1 - y_2 - y_3)$$

$$(x_1, y_1) \oplus ((x_2, y_2) \oplus (x_3, y_3)) = (x_1 - x_2 + x_3, y_1 - y_2 + y_3)$$

Not equal in general. Fails.

- **Additive identity:** We would need (a, b) such that

$$(x, y) \oplus (a, b) = (x, y) \quad \forall (x, y),$$

which fails because the opposite side depends on (x, y) . Fails.

- **Additive inverses:** Not well-defined because there is no true additive identity. Fails.
- **Commutativity of addition:**

$$(x_1, y_1) \oplus (x_2, y_2) = (x_1 - x_2, y_1 - y_2) \neq (x_2 - x_1, y_2 - y_1) = (x_2, y_2) \oplus (x_1, y_1)$$

in general. Fails.

- **Closure under scalar multiplication:** For $c \in \mathbb{R}$,

$$c \odot (x, y) = (-cx, -cy) \in \mathbb{R}^2.$$

Satisfied.

- **Associativity of scalar multiplication:**

$$c \odot (d \odot (x, y)) = c \odot (-dx, -dy) = (cdx, cdy) \neq (-(cd)x, -(cd)y) = (cd) \odot (x, y)$$

Fails.

- **Identity element of scalar multiplication:**

$$1 \odot (x, y) = (-x, -y) \neq (x, y) \text{ in general.}$$

Fails.

- **Distributivity of scalars over vector addition:**

$$c \odot ((x_1, y_1) \oplus (x_2, y_2)) = c \odot (x_1 - x_2, y_1 - y_2) = (-c(x_1 - x_2), -c(y_1 - y_2))$$

$$c \odot (x_1, y_1) \oplus c \odot (x_2, y_2) = (-cx, -cy) \oplus (-cx_2, -cy_2) = (-cx + cx_2, -cy + cy_2)$$

Satisfied.

- **Distributivity of scalar sums over vectors:**

$$(c + d) \odot (x, y) = (-(c + d)x, -(c + d)y) \neq c \odot (x, y) \oplus d \odot (x, y) = (-cx + dx, -cy + dy)$$

Fails.

Conclusion: Only closure under $\oplus$ and $\odot$ and distributivity of scalars over vector addition are satisfied. All other vector space axioms fail, so $(\mathbb{R}^2, \oplus, \odot)$ is **not a vector space over $\mathbb{R}$** .

- On $\mathbb{R}^2$, define two operations

$$(x_1, y_1) \oplus (x_2, y_2) = (x_1 + x_2, 0),$$

$$c \odot (x, y) = (cx, 0)$$

where $c \in \mathbb{R}$. Is $\mathbb{R}^2$, with these operations, a vector space over $\mathbb{R}$?

Solutions: We check the vector space axioms:

- **Closure under addition:**

$$(x_1, y_1) \oplus (x_2, y_2) = (x_1 + x_2, 0) \in \mathbb{R}^2.$$

Satisfied.

- **Associativity of addition:**

$$((x_1, y_1) \oplus (x_2, y_2)) \oplus (x_3, y_3) = (x_1 + x_2 + x_3, 0)$$

$$(x_1, y_1) \oplus ((x_2, y_2) \oplus (x_3, y_3)) = (x_1 + x_2 + x_3, 0).$$

Satisfied.

- **Additive identity:**

$$(x, y) \oplus (0_x, 0_y) = (x + 0_x, 0) = (x, 0) \neq (x, y) \text{ in general} \quad \text{Fails.}$$

- **Additive inverses:** No additive identity exists, so inverses Fail.
- **Commutativity of addition:**

$$(x_1, y_1) \oplus (x_2, y_2) = (x_2, y_2) \oplus (x_1, y_1).$$

Satisfied.

- **Closure under scalar multiplication:**

$$c \odot (x, y) = (cx, 0) \in \mathbb{R}^2.$$

Satisfied.

- **Associativity of scalar multiplication:**

$$c \odot (d \odot (x, y)) = c \odot (dx, 0) = (cdx, 0) = (cd) \odot (x, y).$$

Satisfied.

- **Identity element of scalar multiplication:**

$$1 \odot (x, y) = (x, 0) \neq (x, y) \text{ in general} \quad \text{Fails.}$$

- **Distributivity of scalars over vector addition:** Check $c \odot ((x_1, y_1) \oplus (x_2, y_2)) = c \odot (x_1 + x_2, 0)$:

$$c \odot ((x_1, y_1) \oplus (x_2, y_2)) = (c(x_1 + x_2), 0).$$

Check $c \odot (x_1, y_1) \oplus c \odot (x_2, y_2)$:

$$c \odot (x_1, y_1) \oplus c \odot (x_2, y_2) = (cx_1, 0) \oplus (cx_2, 0) = (cx_1 + cx_2, 0).$$

Since $(c(x_1 + x_2), 0) = (cx_1 + cx_2, 0)$, the axiom holds.

- **Distributivity of scalar sums over vectors:** Check $(c + d) \odot (x, y)$:

$$(c + d) \odot (x, y) = ((c + d)x, 0) = (cx + dx, 0).$$

Check $c \odot (x, y) \oplus d \odot (x, y)$:

$$c \odot (x, y) \oplus d \odot (x, y) = (cx, 0) \oplus (dx, 0) = (cx + dx, 0).$$

Since both sides are equal, the axiom holds.

Conclusion: Only some axioms hold. Since there is no additive identity and inverses fail, $(\mathbb{R}^2, \oplus, \odot)$ is **not a vector space over $\mathbb{R}$** .